FUNCTIONAL SPECIFICATION

VSUART

REVISION HISTORY			
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1. Overview

1.1 Functional overview

This document describes the specification of The Vietnamese Systemware Universal Asynchronous Receiver and Transmitter (UART) IP Core. The hardware component is designed to provide reliable serial communication capabilities for embedded systems and System-on-Chip (SoC) designs. And it also implements standard UART functionality with configurable features to meet diverse application requirements.

Because the design prioritizes adaptability, it provides an AMBA APB Slave Bus interface for system connectivity. All features of the VSUART can be fully configured through control registers.

These features include:

AMBA APB4 interface allows integration into AMBA SoC implementations

- Two clock domains for APB bus and UART Core.
- APB: Support countinous R/W transaction.
- Support 8 or 9-bit serial data mode.
- Functionality based on the 16550-industry standard:
    - Provide selectable parity mode (even, odd or no parity options).
    - One or two Stop bits.
    - 8 data bits.
- Line break generation and detection.
- Transmit and receive FIFO depths up to 128.
- Provides interrupt identification.
- Programmable oversampling rate of 16 or 4.
- Programmable serial data baud rate with 16-bit prescaler Baud Rate Generator.

1.2 Architecture Overview

The picture below presents a simplified block diagram of the UART, highlighting its main hardware components.

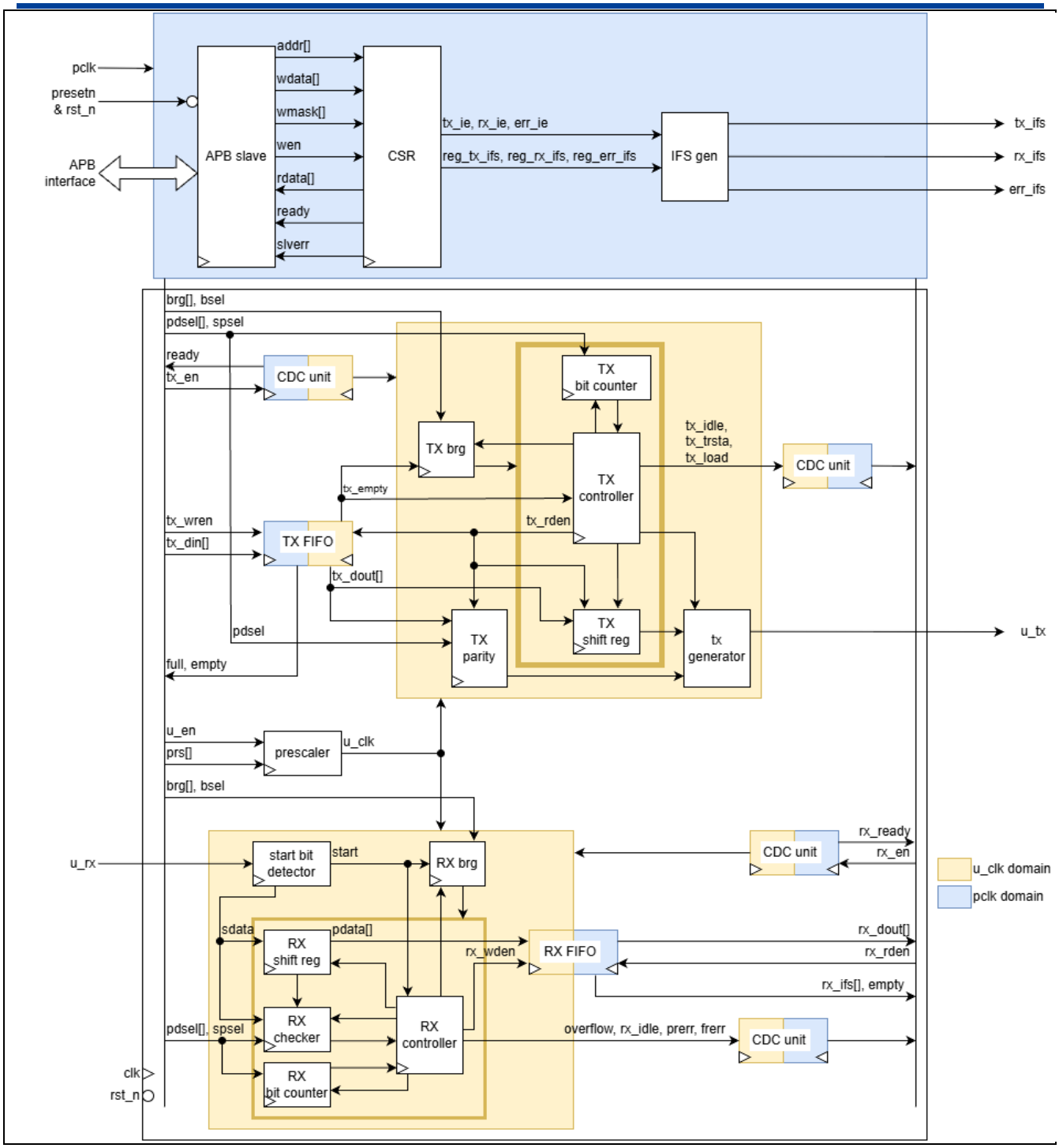


Figure 1-1: Functional Block Diagram of the UART IP

Block	Description
APB subsbystem	APB Slave Interface to connects between UART and system bus.
├ APB slave	Receive APB signal and generate control signal for Uart Tx and Rx
├ CSR (control-status registers)	Contain configuration and status registers
└ Interrupt Controller	Generate interrupt signal based on Uart Tx and Rx status
TX CORE	Tx core handle transmit data path of the UART

├ TX baudrate generator	Generate the baud clock for transmitting by dividing the APB clock using the programmed brg_divisor value.
└ TX	Main block for controlling data flow from FIFO to serial tx output
├ TX FIFO	Buffer memory that stores transmit data before it is processed and sent out serially.
├ TX bit counter	Count transmitted bits (start, data, parity, stop) for frame control.
├ TX controller	FSM control transmitting process
├ TX shift register	Convert parallel data to serial form for transmission.
├ TX parity	Generate parity bits
└ TX generator	Generate final serial signal (tx) combining start, data, parity, and stop bits.
RX CORE	Rx core handle transmit data path of the UART
├ RX Baudrate Generator	Generate the baud clock for UART receive by dividing the APB clock using the baudrate divisor and oversampling value.
├ RX Start Detector	Detects falling edge indicating start bit of a transmission on rx line.
└ RX	Main block for controlling data receive from rx pin
├ RX FIFO	Stores received data.
├ RX bit counter	Counts received bits to determine end of frame.
├ RX controller	FSM control receiving process
├ RX shift register	Convert incoming serial bits to parallel data format.
└ RX checker	Check errors on reception path (parity, framing errors).

Table 1-1: Block Description of the UART IP

2. Interfaces description

2.1 System source

Name	Width	I/O	Description	Init.
sys_clk	1	I	System Clock (40 ~ 200 MHz)	-
rst_n	1	I	Asynchronous reset (active '0') If this system reset is activated: • System status is reset (included apb and uart subsystem) • Registers are reset to initial value	-

Table 2-1: System source interface signal descriptions

2.2 APB slave interface

Name	Width	I/O	Description	Init.
PCLK	1	I	Rising edge clock (max: 72 MHz) All APB signals are timed against the rising edge of PCLK.	-
PRESETn	1	I	Asynchronous reset (active '0'). The active low PRESETn is for APB slave driver reset	-
PADDR	16	I	PADDR is the APB address bus. The address +FFFFh is dedicated to transmit data access (write data to the TX FIFO), while all other addresses are used to access configuration and status registers in the register bank.	-
PSEL	1	I	APB select signal. Asserts HIGH to select the UART slave during a data transfer.	-
PENABLE	1	I	APB enable signal. Asserts HIGH to indicate the access phase of APB transfer.	-
PWRITE	1	I	APB write signal. PWRITE indicates an APB write access when HIGH and an APB read access when LOW.	-
PWDATA	32	I	APB write data. Carries 32 bit data from the APB master to the UART IP	-
PSTRB	4	I	APB write strobe signals. Indicates which byte lanes to update during a write transfer. Each bit corresponds to one byte of PWDATA and indicates which bytes are valid during a write transaction.	-
PREADY	1	O	APB ready signal. Asserts HIGH after the UART slave completes the current transfer and is ready to proceed.	1
PRDATA	32	O	APB read data.	0
PSLVERR	1	O	APB Transfer error signal. Asserts HIGH to indicate an error condition.	0

			Condition: 1. To write into SPSEL, BSEL, PDSEL, VSUART_BRG, the TX core and RX core must be disabled first 2. To write into VSUART_PRS, the UEN[2] must be 0 first 3. Try to read data when RX FIFO is empty	

Table 2-2: APB Slave interface signal description

2.3 UART interface

Name	Width	I/O	Description	Init.
rx	1	I	UART receive data input	-
tx	1	O	UART transmit data output.	1

Table 2-3: Uart interface signal description

2.4 Interrupt Flag Status (IFS)

All of following signals are active high level.

Name	Width	I/O	Description	Init.
ifs_rx	1	O	RX interrupt flag	0
ifs_tx	1	O	TX interrupt flag	0
ifs_err	1	O	Error interrupt flag. Asserted when a reception error occurs (parity error, frame error)	0

Table 2-4: Interrupt Flag Status signal description

3. Control Registers

3.1 Register summary

Name	Register Offset	POR	Description	Type
VSUART_MODE	'h0000	'h0001_0000	UART mode register. - Controls frame format including data length, parity, stop bits. - Controls baud rate oversampling value. - Enable Uart Tx, Rx.	R/RW
VSUART_CR	'h0004	'h0000_0000	UART control register. Control stransmit and receive interrupt mode selection.	R/RW
VSUART_STA	'h0008	'h0000_0024	UART status register. Provides status including busy/idle information, FIFO conditions (empty, full, overflow) of the transmitter, receiver, and detected error flags.	R/R1C
VSUART_PRS	'h000C	'h0000_0000	UART prescaler	R/RW
VSUART_BRG	'h0010	'h0000_0001	UART Baud Rate Generator Register. Contains value for baudrate divisor value	R/RW
VSUART_IE	'h0014	'h0000_0000	UART Interrupt enable register. Enables interrupt sources of TX, RX, and error interrupt.	R/RW
VSUART_IFS	'h0018	'h0000_0000	UART Interrupt flag register. Contains current active interrupt status (TX, RX and error interrupt).	R/W1C

Table 3-1: Register summary description

Type of register:

- R: Read only
- WR: Write-read
- W1C: Write 1 to Clear
- R1C: Read 1 to Clear

3.2 UART mode register (VSUART_MODE)

Name	Bit map	Type	Initial value	Description
UEN	[2:0]	RW	3'b000	UART enable • bit [0]: enable UART TX 1: UART TX is enabled (if UEN[2] = 1) 0: UART TX is disabled • bit [1]: enable UART RX 1: UART RX is enabled (if UEN[2] = 1) 0: UART RX is disabled • bit [2]: enable UART core (TX, RX, Prescaler)
-	[7:3]	R	5'b00000	Reserved
BSEL	[8]	RW	1'b0	Baud rate oversample selection • 1 : 4x baud clock (Oversample = 4) • 0 : 16x baud clock (Oversample = 16)
-	[15:9]	R	7'b0000000	Reserved

PDSEL	[17:16]	RW	2'b01	Parity and data mode selection • 11b : 9-bit data, no parity • 10b : 8-bit data, even parity • 01b : 8-bit data, odd parity • 00b : 8-bit data, no parity
-	[23:18]	R	6'b000000	Reserved
SPSEL	[24]	RW	1'b0	Stop bit selection • 1: 2 stop bits • 0: 1 stop bit
-	[31:25]	R	7'b0000000	Reserved

Table 3-2: Uart mode register description

Note: Users only can access BSEL, PDSEL, SPSEL when disabled (UEN[2] = 0 or UEN[1:0] = 0). If writing in enabled state, PSLVERR will be asserted and this writing is ignored.

BSEL:

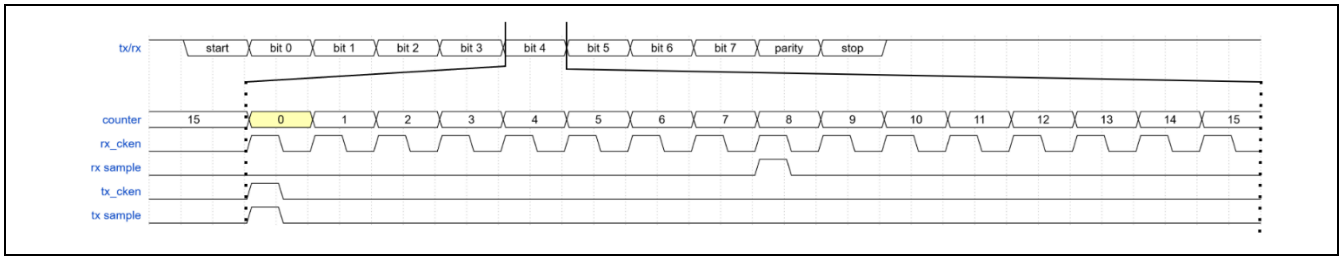


Figure 3-1: Diagram of 16x baud clock

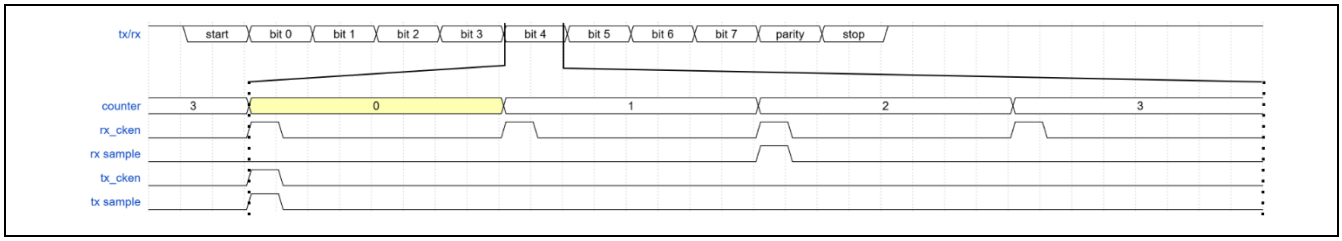


Figure 3-2: Diagram of 4x baud clock

3.3 UART control register (VSUART_CR)

Name	Bit map	Type	Initial value	Description
TXISEL	[1:0]	RW	2'b00	Transmit Interrupt Mode Selection •11b: Interrupt is generated when transmit buffer is overflow •10b: Interrupt is generated when the character is transferred to the Transmit Shift register, and the transmit buffer (tx fifo) becomes empty. •01b: Interrupt is generated when all characters are transmitted. This implies that all transmitted operations are completed. •00b: Interrupt is generated when a character is transferred from the tx fifo to the Transmit Shift register
-	[7:2]	R	6'b000000	Reversed
RXISEL	[9:8]	RW	2'b00	Receive Interrupt Mode Selection • 11b: Interrupt flag bit is set when receive buffer is full • 10b: Interrupt flag bit is set when receive buffer is 3/4 full • 0xb: Interrupt flag bit is set when a character is received (buffer not empty)
-	[31:10]	R	22'b0	Reversed

Table 3-3: Uart control register description

Note: accessing these above fields only when idle state (TISTA = 1 and RISTA = 1)

3.4 UART Status register (VSUART_STA)

Name	Bit map	Type	Initial value	Description
TBSTA	[0]	R	1'b0	Transmit Buffer Full Status • 1: Transmit buffer is full • 0: Transmit buffer is not full
TRSTA	[1]	R	1'b0	Transmit Shift Register is Empty • 1: Transmit shift register is empty and transmit buffer is empty (the last transmission has completed) • 0: Transmit shift register is not empty, a transmission is in progress or queued in the transmit buffer
TISTA	[2]	R	1'b1	Transmit IDLE Status • 1: Transmit is IDLE • 0: Data is being transmitted
FRERR	[3]	R1C	1'b0	Receive Framing Error Status • 1: Framing error has been detected for the current character • 0: Framing error has not been detected This bit is cleared as soon as the VSUART_STA is read.
PRERR	[4]	R1C	1'b0	Receive Parity Error Status • 1: Parity error has been detected for the current character • 0: Parity error has not been detected This bit is cleared as soon as the VSUART_STA is read.

RISTA	[5]	R	1'b1	Receive IDLE Status • 1: Receiver is IDLE • 0: Data is being received
RBOSTA	[6]	R1C	1'b0	Receive Buffer Overflow Status • 1: Receive buffer has overflowed • 0: Receive buffer has not overflowed This bit is cleared as soon as the VSUART_STA is read.
RBDSTA	[7]	R	1'b0	Receive Buffer Data Available Status • 1: Receive buffer has data • 0: Receive buffer is empty
-	[31:8]	R	24'b0	Reversed

Table 3-3: Uart status register description

Read-1-to-clear type means if the asserting bit map is read, this bit will be deasserted to 0 intermediately.

The priority of "Reading 1" from APB is higher than internal status signals.

3.5 UART prescaler register (VSUART_PRS)

Name	Bit map	Type	Initial value	Description
PRS	[3:0]	RW	4'b0	Scale-down divisor value
-	[31:4]	R	28'b0	Reserved

Table 3-8: Uart prescaler register description

Note: Users only can access PRS when disabled (UEN[2] = 0). If writing in enabled state, PSLVERR will be asserted and this writing is ignored.

This register is for scaling down system clock (sys_clk) for UART core operating clock (u_clk)

The condition to change PRS bits is that the prescaler block must be disabled, and there is no generated clock (UEN[2] = 0). Use the following equation:

$$F_{u_clk} = \frac{F_{sys_clk}}{2 * (PRS + 1)}$$

Example timing diagram:

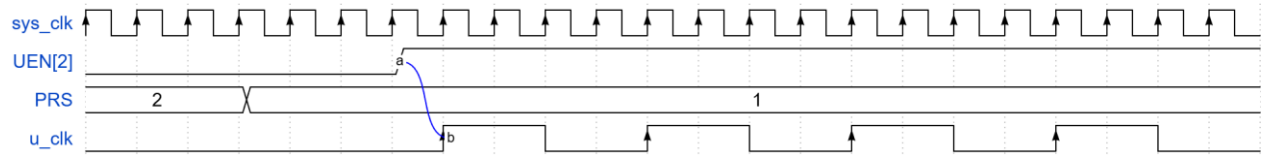


Figure 3-3: Example diagram of setting PRS ($F_{sys_clk} = 4F_{u_clk}$)

3.6 UART Baud Rate Generator register (VSUART_BRG)

Name	Bit map	Type	Initial value	Description
BRG	[15:0]	RW	16'b1	Baud Rate Divisor value
-	[31:16]	R	16'b0	Reserved

Table 3-4: Uart Baudrate Generator register description.

Note: Users only can access BRG when disabled (UEN[2] = 0 or UEN[1:0] = 0). If writing in enabled state, PSLVERR will be asserted and this writing is ignored.

BRG calculation

To set the desired Baudrate, first we need to set the value of VSUART_**BRG** and oversample **VSUART_MODE.BSEL** register value.

Use the following equation:

$$\text{UART_BRG} = \frac{\text{F}_{\text{u_clk}}}{\text{Oversample} \times \text{BaudRate}_{\text{desired}}} - 1$$

Next, based on the VSUART_BRG register value, we have the actual Baud Rate

$$\text{BaudRate}_{actual} = \frac{\text{F}_{\text{u_clk}}}{\text{Oversample} \times (\text{UART_BRG} + 1)}$$

The TX bit rate is equal to the programmed baud rate:

$$\text{Freq}_{\text{tx}} = \text{BaudRate}_{\text{actual}} = \frac{\text{F}_{\text{u_clk}}}{\text{Oversample} \times (\text{UART_BRG} + 1)}$$

The receiver oversamples the incoming serial data based on the configured oversampling factor.

Accordingly, the RX sampling frequency is:

$$\text{Freq}_{\text{rx}} = \text{BaudRate}_{\text{actual}} \times \text{Oversample} = \frac{\text{F}_{\text{u_clk}}}{(\text{UART_BRG} + 1)}$$

The following **example** will help you understand more initial condition:

- F_{sys_clk} = 40 MHz
- PRS = 4.
- Desired Baurate = 9600
- Oversample = 4 (configured by BSEL = 1)

Calculate clk for UART core:

$$F_{u_clk} = \frac{F_{sys_clk}}{2 * (PRS + 1)} = \frac{40}{2 * (4 + 1)} = 4\ MHz$$

Calculate VSUART_BRG Register and desired Baudrate:

$$\text{UART_BRG} = \frac{\text{F}_{\text{u_clk}}}{\text{Oversample} \times \text{BaudRate}_{\text{desired}}} - 1 = \frac{4{,}000{,}000}{4 \times 9600} - 1 = 103$$

$$\text{BaudRate}_{\text{actual}} = \frac{4.000.000}{4 \times (103 + 1)} = 9615(\text{bits/s})$$

$$\text{Freq}_{\text{tx}} = \text{BaudRate}_{\text{actual}} = \frac{4.000.000}{4 \times (103 + 1)} = 9615(Hz)$$

$$\text{Freq}_{\text{rx}} = \frac{4{,}000{,}000}{103 + 1} = 38461(\text{Hz})$$

Error desired Baudrate:

$$\text{Error} = \frac{9615 - 9600}{9600} = 0.16\%$$

Note: The VSUART_BRG register supports values from 1 to 65535. A value of 0 is not allowed, as it would result in an invalid RX sampling pulse.

With system clock: 112 MHz, PRS = 1

$$F_{u_clk} = \frac{F_{sys_clk}}{2 * (PRS + 1)} = \frac{112}{2 * (1 + 1)} = 28\ MHz$$

Target Baud Rate	BRG value	BSEL value	Actual Baud Rate	% Error	Note
2400	728	0	2400.55	0.02%	-
	2916	1	2399.73	-0.01%	-
4800	364	0	4794.52	-0.11%	-
	1457	1	4801.1	0.02%	-
9600	181	0	9615.38	0.16%	-
	728	1	9602.19	0.02%	-
19.2 K	90	0	19230.77	0.16%	-
	364	1	19178.08	-0.11%	-
38.4 K	45	0	38043.48	-0.93%	-
	181	1	38461.54	0.16%	-
76.8 K	22	0	76086.96	-0.93%	-
	90	1	76923.08	0.16%	-
115.2 K	14	0	116666.67	1.27%	-
	60	1	114754.1	-0.39%	-
Min Rate	65535	0	26.7	-	-
	65535	1	106.81	-	-
Max Rate	1	0	875 K	-	-
	1	1	3500 K	-	-

Table 3-5: Baudrate table for 112MHz system clock and PRS = 1

3.7 UART Interrupt Enable register (VSUART_IE)

Name	Bit map	Type	Initial value	Description
TXTE	[0]	RW	1'b0	Transmitter Interrupt Enable Bit. Enable TXIFC
-	[7:1]	R	7'b0000000	Reversed
RXTE	[8]	RW	1'b0	Receiver Interrupt Enable bit. Enable RXIFC
-	[15:9]	R	7'b0000000	Reversed
ERRIE	[16]	RW	1'b0	Error Interrupt Enable bit Enable ERRIFC
-	[31:17]	R	15'b0	Reversed

Table 3-6: Uart Interrupt Enable register description

Note: accessing these above fields only when idle state (TISTA = 1 and RISTA = 1)

3.8 UART Interrupt Flag Status register (VSUART_IFS)

Name	Bit map	Type	Initial value	Description
TXIFS	[0]	W1C	1'b0	Transmitter Interrupt Flag Status
-	[7:1]	R	7'b0	Reversed
RXIFS	[8]	W1C	1'b0	Receiver Interrupt Flag Status
-	[15:9]	R	7'b0	Reversed
ERRIFS	[16]	W1C	1'b0	Error Interrupt Flag Status
-	[31:17]	R	15'b0	Reversed

Table 3-7: Uart Interrupt Flag Status register description

This Write-1-to-clear register when being asserted can be cleared to 0 by writing 1 to relatively bit map.

The priority of "Writing 1" from APB is higher than internal interrupt signals.

4. Usage

4.1 Enabling UART

The UART module is enabled by setting the bits UEN.

- UEN[0] is for enabling UART TX
- UEN[1] is for enabling UART RX
- UEN[2] is for enabling UART core (TX, RX, prescaler)

Note:

- Enable prescaler block: UEN[2] = 1
- Enable UART TX: UEN[2] = 1 and UEN[0] = 1
- Enable UART RX: UEN[2] = 1 and UEN[1] = 1
- When UART TX is disabled, any pending transmission and transmitting is aborted and transmission buffer is reset.
- When UART RX is disabled, receiption buffer is reset.

4.2 Setting Prescaler and data frame format

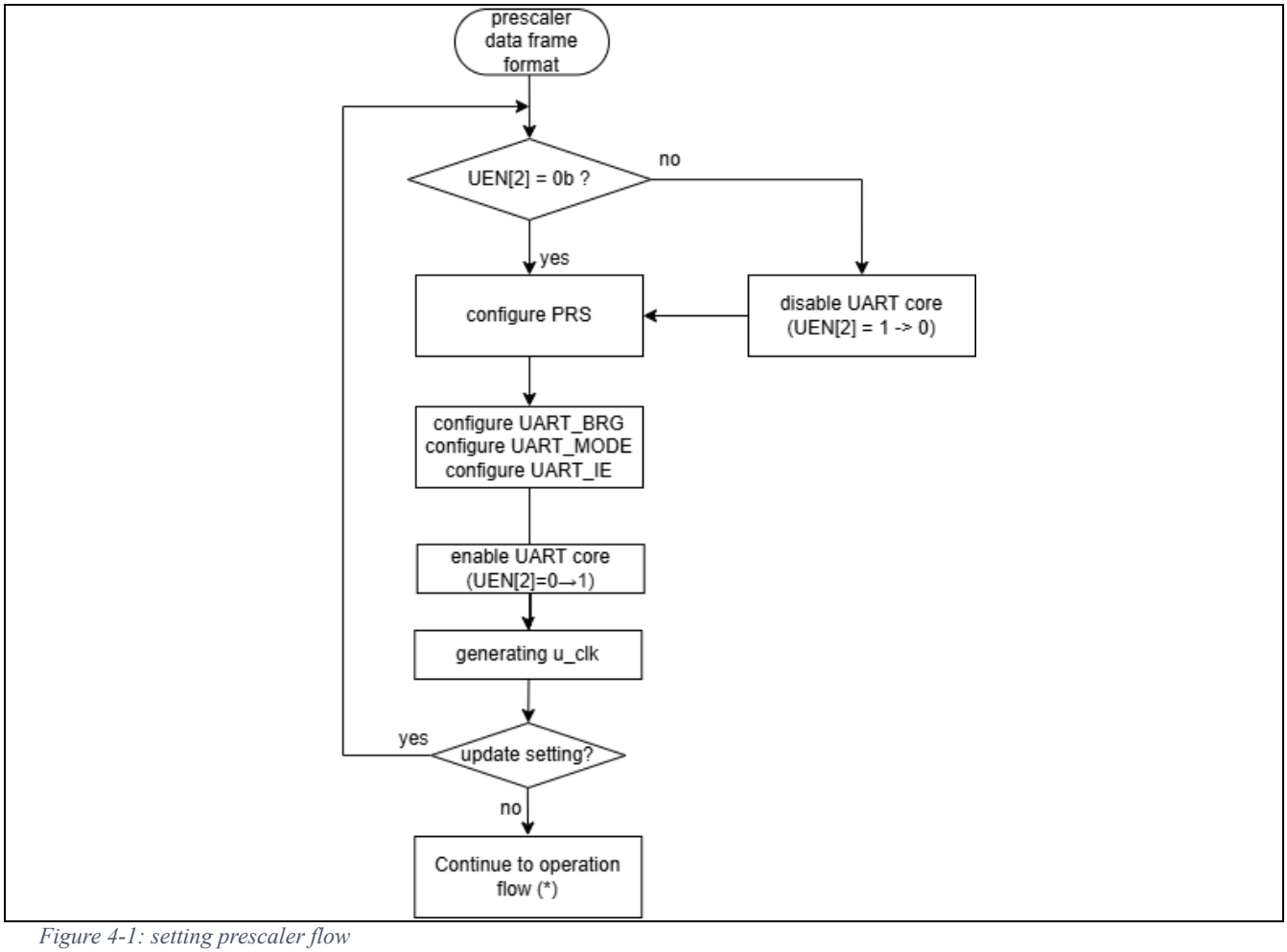


Figure 4-1: setting prescaler flow

(*): Refer section 4.4 for setting operation flow.

4.2.1 Setting Prescaler

Refer section 3.5 for setting scale-down divisor **PRS.**

4.2.2 Setting Baudrate

Refer section 3.2 for setting oversampling **BSEL.**

Refer section 3.6 for setting baudrate divisor **BRG.**

TX baudrate generation

For tx_cken generation, UEN[0] must be set to 1 (UART TX enable)

tx_cken will be generated while transmitting data (tx status != IDLE). Meanwhile, while tx status = IDLE, system will stop generating tx_cken.

RX baudrate generation

For rx_cken generation, UEN[1] must be set to 1 (UART RX enable)

rx_cken will be generated while receiving data (rx status != IDLE). Meanwhile, while rx status = IDLE, system will stop generating tx_cken.

4.2.3 Setting parity-data, stop-bit

Refer section 3.2 for setting parity-data mode **PDSEL.**

Refer section 3.2 for setting stop-bit mode **SPSEL.**

4.3 Error detection

No error detection in TX.

RX can detect parity error, and frame error during receiving.

4.3.1 Frame error detection

If a '0' is detected (error condition) during reception of the stop bit, FRERR shall be set (=1). The Error Interrupt Flag Status (ERRIFS) shall be asserted.

FRERR shall remain set until it is cleared by reading the VSUART_STA register through APB.

4.3.2 Parity error detection

If received parity bit is invalid (a value different from the specified configuration), PRERR shall be set (=1). The Error Interrupt Flag Status (ERRIFS) shall be asserted.

PRERR shall remain set until it is cleared by reading the VSUART_STA register through APB.

4.3.3 Interrupt control

There are 2 interrupt signals for operation, and 1 interrupt signal for error.

Refer section 3.3 for selecting interrupt operation mode TX and RX**.**

Signals	Flag	Flag enable	Note
ifs_rx	RXIFS	RXIE	Select mode via RXISEL
ifs_tx	TXIFS	TXIE	Select mode via TXISEL

ifs_err	ERRIFS	ERRIE	FRERR & PRERR

Table 4-1: List of interrupt signals

Interrupt Control

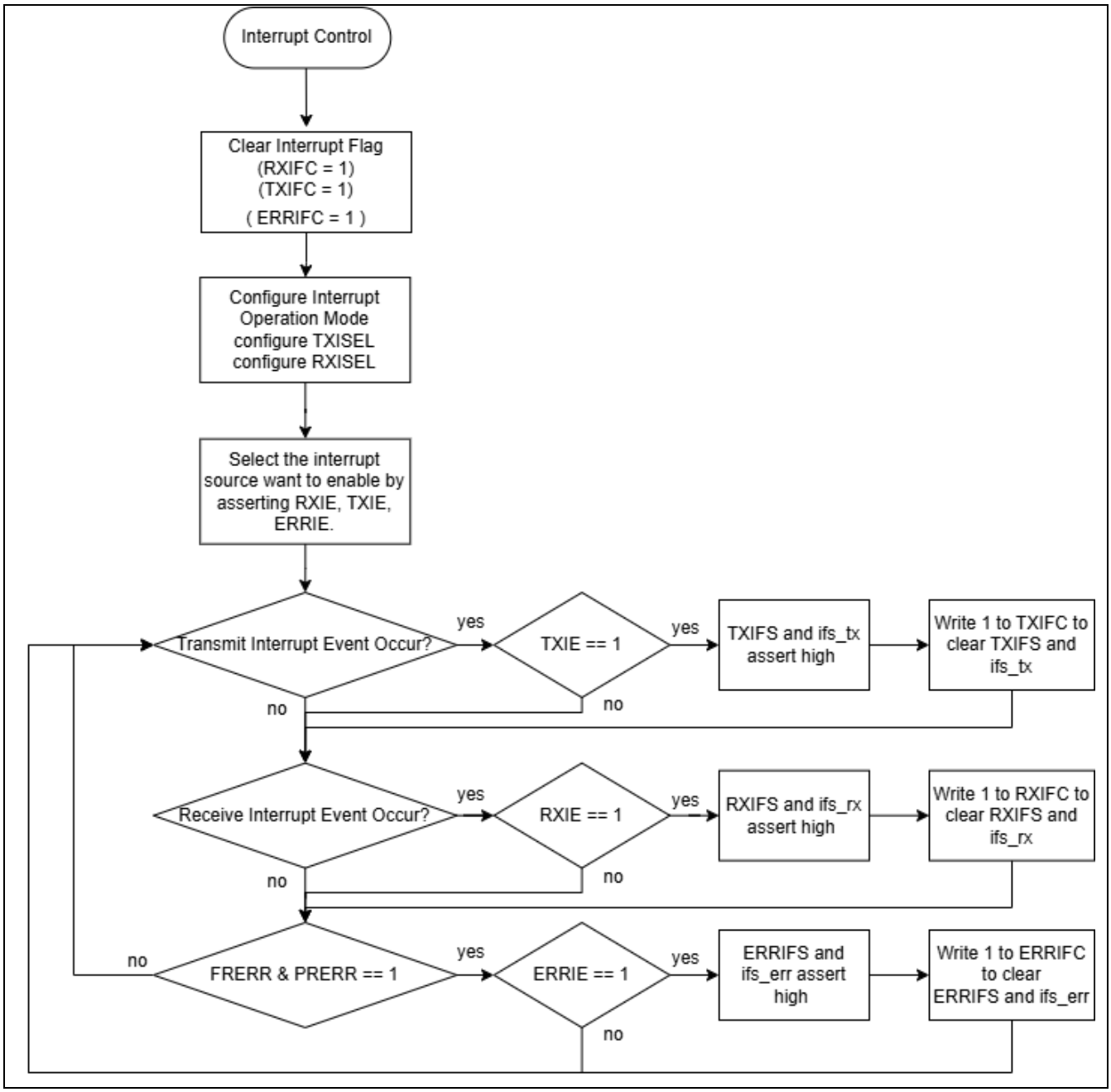


Figure 4-2: Interrupt Operation flow

4.4 Operation flow

4.4.1 Charater transmission

(1). Configure Baudrate, data frame format, TX interrupt mode

(2). Enable UART TX (UEN[0] = 1 & UEN[2] = 1)

(3). Write data to TX FIFO through APB is similar with APB write transaction with:

- PADDR[15 : 0] = FFFFh

- PWDATA[8 : 0] = Written data. PWDATA[31 : 9] reserved.
- PSTRB[3:0] = 0011b
- PWRITE = 1

For continuous transaction, view **Figure 4-4** for details

(4). While TX FIFO is not empty, TX Core shall get data from TX then operate UART transmission.

Note:

- When TX is disabled, the Tx FIFO operation is stopped; therefore, writing to the Tx FIFO is not allowed. Therefore, in the following sequence:

UEN[0] = 0 → Tx FIFO Write (CPU) → UEN[0] = 1 → No action (Tx FIFO remains empty)

This sequence assumes that no data is written to the Tx FIFO while UEN[0] = 0, and therefore the Tx FIFO remains empty after UEN[0] is set back to 1.

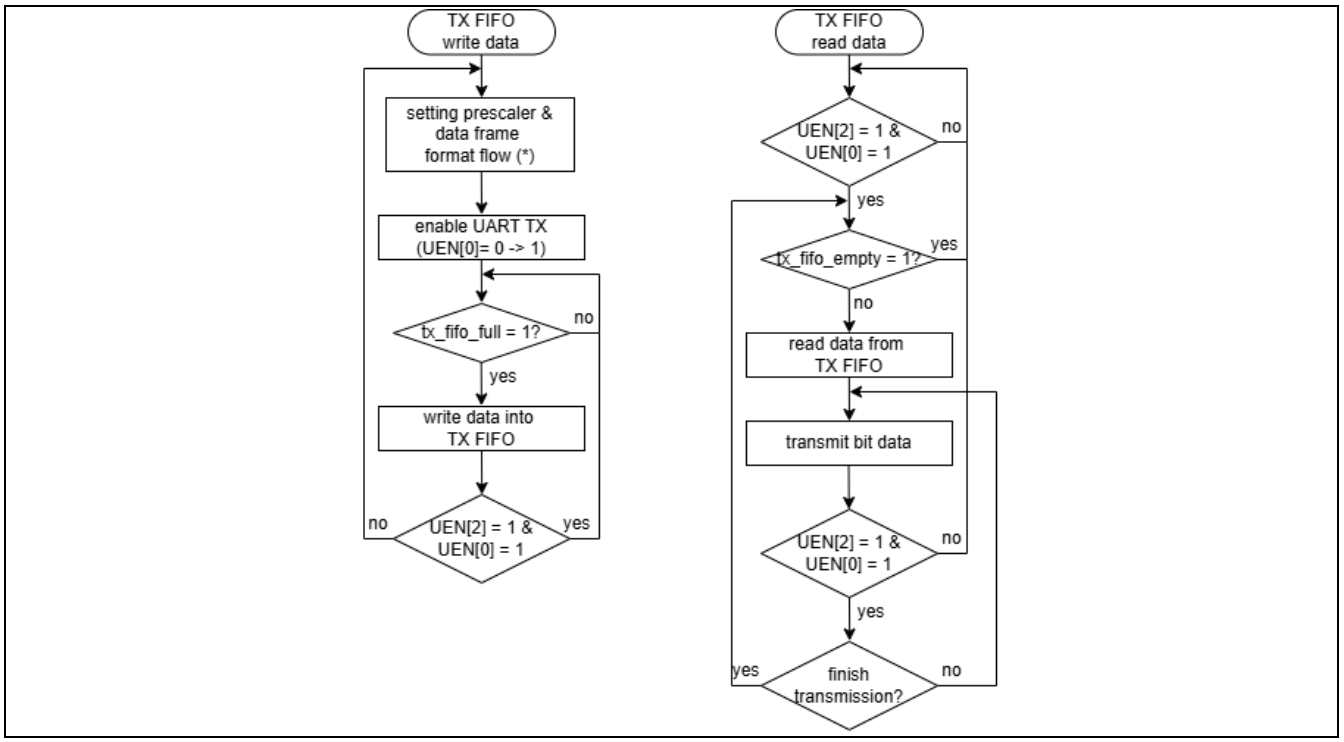


Figure 4-3: TX FIFO writing and reading data flow

(*): refer to section 4.2

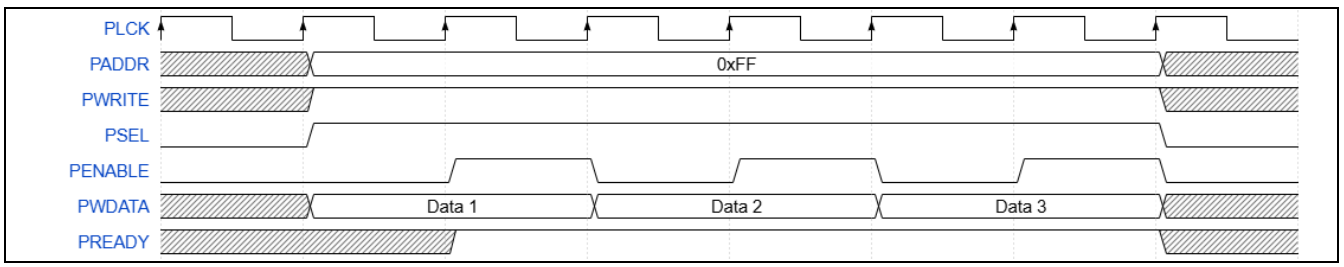


Figure 4-4: APB continuous FIFO writing transaction

4.4.2 Charater reception

(1). Set up RX enable (UEN[1] = 1)

(2). Set up Baudrate, data frame format, RX interrupt mode

(3). Receiving data from RX FIFO through APB is similar with APB read transaction with:

- PADDR[15 : 0] = FFFFh
- PRDATA[8 : 0] = read data. PWDATA[31 : 9] reserved.
- PWRITE = 0

For continuous transaction, view for details

(4). If RX FIFO is empty but getting data through APB, this will be read transaction with error response.

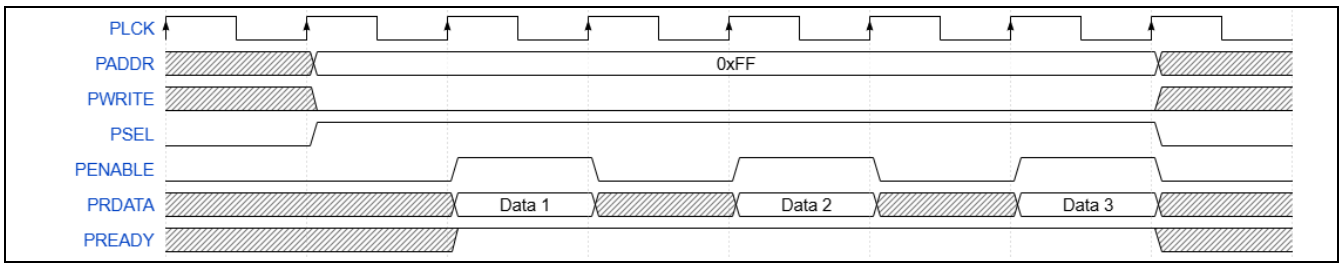


Figure 4-5: APB continuous FIFO reading transaction

5. Cautions

- This module has frequency constraint: $F_{sys_clk} > F_{pclk} > F_{u_clk}$